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LABELS AND ANNOTATIONS

MATLAB Drive

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5G AND 6G LAB

LAB EXPERIMENT

New Folder

ASSEMENT QNO 1.m

EX: 2.m

ex 1.m

EX 2 .m

EX 2..m

Ex 2.m

ex 5.m

ex1.m

Workspace

Name	Value	Size	Class
AF	1×361 double	1×361	double
k	361	1×1	double
N	8	1×1	double
phase	[0;1.5708;3.1...	8×1	double
theta	1×361 double	1×361	double
theta0	30	1×1	double

EX1.m

ex2.m

ex3.m

EX4.m

untitled4.m

EX6.m

EX7.m

Figure 1

```

9 % Steering Vector
10 AF = zeros(size(theta));
11
12 for k = 1:length(theta)
13
14     phase = pi*(0:N-1)*(sind(theta(k)) - sind(theta0));
15     AF(k) = abs(sum(exp(1j*phase)));
16
17 end
18
19 % Normalize Array Factor
20 AF = AF/max(AF);
21
22 % Plot Beam Pattern
23 figure;
24 plot(theta,20*log10(AF),'LineWidth',2);
25 grid on;
26 axis([-90 90 -40 0]);
27
28 xlabel('Angle (Degrees)');
29

```

Command Window

>>

Analog Beamforming Radiation Pattern